Figure 12: Database Schema Diagram

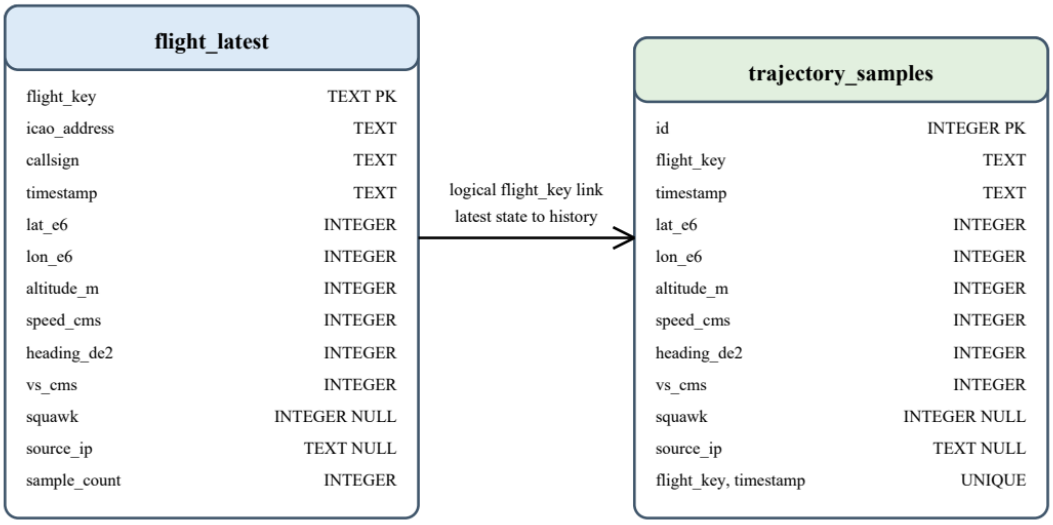


Figure 12: Database Schema Diagram

Table 4: Database Tables

Table	Purpose
flight_latest	Stores the latest known state for each flight key.
trajectory_samples	Stores historical position and movement samples over time.

4.4 Main System Modules

Table 3: Main System Modules

Module	File	Responsibility
Flask application	app.py	Dashboard routes, JSON APIs, status and exports.
Data source	data_source.py	Live/replay selection, configuration and receiver state.
Raw decoder	preprocessing/pingstation_raw_decoder.py	Frame splitting, field decoding and aircraft-state merging.
Preprocessor	preprocessing/pingstation_preprocess.py	Unit normalization and identity/timestamp cleanup.
Storage	storage/flight_data_store.py	SQLite, trajectory history, prediction and file exports.
Dashboard	templates/index.html	Map, list, details, route trace and altitude chart.